BrainImation: A Browser-Based Teaching Environment for Learning Creative Coding with Live EEG, Optical PPG/fNIRS, and Head Motion Sensors

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Live Tool: https://kylemath.github.io/Brainimation/
Source Repository: https://github.com/kylemath/Brainimation

Article Type: Full Length Article (Research Article / New Method)

Keywords (6): neuroscience education; electroencephalography (EEG); functional near-infrared spectroscopy (fNIRS); photoplethysmography (PPG); brain–computer interface (BCI); event-related potential (ERP)

Highlights (≤ 85 characters per bullet point including spaces):

- Browser-based coding tool for teaching EEG, optical PPG/fNIRS, and head motion. (79 chars)
- Connects Muse 2016, Muse 2, Muse S, and Muse S Athena with no software install. (79 chars)
- Built-in simulator lets students write and test code at home without hardware. (78 chars)
- Visual Catalogue of 226 example and student sketches across 18 categories. (74 chars)
- Evaluated on 69 neurofeedback and ERP projects from 30 students ($p < 10^{-11}$). (77 chars)

Declarations

Ethics and SAGER Statement: Classroom code artifacts from `PSYCH 403` were analyzed in de-identified form in accordance with procedures approved by the University of Alberta Research Ethics Board (Protocol `Pro00050069`). Because software source files were analyzed in de-identified archival form without demographic linkage, participant sex and gender dimensions were not recorded and do not affect the deterministic signal-processing benchmarks.

Data and Code Availability: The live BrainImation environment is available at https://kylemath.github.io/Brainimation/. Source code (`muse-browser.js`, `index.html`, and all 226

catalogue sketches and thumbnails in `examples/`) and analysis scripts are archived at https://github.com/kylemath/Brainimation and https://github.com/kylemath/eegeduPaper.

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Declaration of Competing Interest: The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement: Kyle E. Mathewson: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Tamari Shalamberidze:** Methodology, Software, Validation, Investigation, Data curation, Writing – review & editing. **Claire J. Scavuzzo:** Conceptualization, Methodology, Investigation, Supervision, Writing – review & editing.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process: During the preparation of this work the authors used large language model coding and editing assistants to verify code-to-manuscript consistency and polish prose formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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